Clinical Guideline for:
Continuous Mandatory Ventilation (CMV)

SUMMARY

This document provides guidance on the use of CMV in the neonatal unit

KEY POINTS

CMV can be used with either pressure controlled or volume targeted ventilation. All breaths are delivered by the ventilator at a set rate regardless of the babies own respiratory effort. Morphine may be considered in babies who appear distressed or are breathing out of synchrony with the ventilator. Pancuronium may also be used for babies who have a high ventilator requirement but should not be used without morphine.

Contents

1. Setting up conventional ventilation 2
2. Adjusting the ventilator during CMV 2
3. Weaning babies from CMV 2
4. Pressure controlled CMV 2
5. MONITORING COMPLIANCE WITH THIS GUIDELINE 3
6. PUBLICATION DETAILS 3

This clinical guideline is designed to provide advice, guidance and direction to staff whilst leaving room for professional judgement, and adaptation, to fit individual circumstances.

1. Setting up conventional ventilation

- The initial inspiratory time (Ti) is set at 0.35-0.4 seconds.
- The positive end expiratory pressure (PEEP) is initially set at 4-6 cms of water
- Set targeted tidal volume at 4-5mls/kg with pressure limit 25-30cm/H_2O (enough to allow the volume to be delivered).
- The rate is initially set at 40-60 breaths per minute.

2. Adjusting the ventilator during CMV

- Low PaO_2
  - Increase FiO_2
  - Increase MAP by increasing Vte (set tidal volume)
  - Increasing PEEP to recruit more alveoli or
  - Increasing inspiratory time(T_i) to a maximum of 0.5s
- High PaO_2
  - Reduce FiO_2
- High PCO_2
  - Increase tidal and/or minute volume by increasing set tidal volume or increase back-up rate
- Low PCO_2
  - Reduce tidal and/or minute volume by reducing set tidal volume or reduce back-up rate.

3. Weaning babies from CMV

- Consider changing to a mode with synchronous ventilation
- Consider the use CPAP via the infant flow driver or optiflow post extubation
- Reduce Vte to 4mls/kg.

4. Pressure controlled CMV

If a pressure controlled mode of CMV is selected start with peak inspiratory pressure (PIP) of 18-20 measuring tidal volume to achieve 4-5mls/kg. Adjust PIP by increments of 2cm H2O and/or respiratory rate according to blood gas measurement.

5. MONITORING COMPLIANCE WITH THIS GUIDELINE

Any concern or non-compliance with this guideline that is identified through the investigation of clinical incidents, complaints or claims will be reviewed as per Trust policy and may result in an audit and / or amendment to the guideline.

6. PUBLICATION DETAILS

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